

Swaham Pitambar Patra

Task 1:

```
1  ### Q1.
2  from abc import ABC, abstractmethod
3  class Shape:
4      @abstractmethod
5      def area(self):
6          pass
7
8  # Create a class Circle that inherits from Shape.
9  class Circle(Shape):
10     def __init__(self, radius):
11         self.radius = radius
12
13     # Implement the area() method to return:
14     def area(self):
15         return 3.14 * self.radius * self.radius
16
17 # Instantiate a Circle object with *radius = 3* and print its area.
18 obj = Circle(3)
19 print(f"Area of Circle = {obj.area()}")
```

PROBLEMS TERMINAL DEBUG CONSOLE PORTS

```
PS C:\Users\Swaham Patra\OneDrive\ITR Python+DS> python -u "c:\Users\Swaham Patra\OneDrive\ITR Python+DS\Tasks
Area of Circle = 28.259999999999998
PS C:\Users\Swaham Patra\OneDrive\ITR Python+DS>
```

Task 2:

```
1  ### Q2.
2  class Animal:
3      def speak(self):
4          print("Animal")
5  # Create a class Bird that inherits from Animal.
6  class Bird(Animal):
7      # Override the speak() method to print:
8      def speak(self):
9          print("Tweet!") # Tweet!
10
11 # Create a Bird object and call the speak() method.
12 obj = Bird()
13 obj.speak()
```

PROBLEMS TERMINAL DEBUG CONSOLE PORTS

```
PS C:\Users\Swaham Patra\OneDrive\ITR Python+DS> python -u "c:\Users\Swaham Patra\OneDr
Tweet!
PS C:\Users\Swaham Patra\OneDrive\ITR Python+DS> |
```

Task 3:

```
1  ### Q3.
2  # Define a class Superman with a method fly().
3  class Superman:
4      def fly(self):
5          print("Flying!")
6
7  def lift_off(obj):
8      obj.fly()
9
10 # Create an object of Superman and pass it to the lift_off() function.
11 obj = Superman()
12
13 lift_off(obj)
```

PROBLEMS TERMINAL DEBUG CONSOLE PORTS

```
PS C:\Users\Swaham Patra\OneDrive\ITR Python+DS> python -u "c:\Users\Swaham Patra\OneDrive\ITR Python+DS\Tasks 10-0
Flying!
PS C:\Users\Swaham Patra\OneDrive\ITR Python+DS> |
```